AI-Powered Collections Strategy

Leveraging Agentic AI for Scalable, Fair, and Effective Debt Management at Geldium

Prepared for Geldium's Head of Collections
Apoorv Bajpai, AI Transformation Consultant, Tata iQ

How the System Works: Four Parts

Inputs
What the system sees

- Customer profile: income, employment, debt-to-income, credit score
- Credit utilization and 6-month payment status
- Cleaned automatically (median fill, standard labels)
- Age and Location used only for fairness checks

Decision Logic
How it decides

- Risk model gives each customer a delinquency probability
- Business rules turn the score into a Low, Medium or High tier
- Top reasons are saved with every score

Actions
What it triggers

- Low: no extra outreach
- Medium: reminder before the due date plus payment-plan information
- High: payment-plan or hardship offer and agent contact

Learning Loop
How it improves

- Track repayment, engagement and cost
- Compare pilot customers with a control group
- Retrain when accuracy or fairness drifts

Outcomes feed back into the inputs and decision rules, so the system improves with every cycle.

How the System Works: Rules by Risk Tier

Tier	Example trigger (illustrative)	Action	Who acts
Low	Utilization under 30%, on-time payments, low model score	No extra outreach; standard statement reminder	Automated
Medium	Utilization 50% or higher, or mid-range model score	Reminder before the due date plus flexible payment-plan information	Automated
High	Top 20% model score, or recent late or missed months	Personalised payment plan or hardship offer; agent call	Agent approves

Thresholds are examples. They must be calibrated on Geldium's data once the model is validated. In our data, delinquency was 12% below 30% utilization and 17-20% above it.

Role of Agentic AI

Autonomous	Human Oversight
Refresh data and score customers daily	Approve payment plans and hardship offers
Send reminders by SMS or email from approved templates	Handle disputes, complaints and exceptions
Choose timing and channel within preset limits	Approve calls, escalation or any step beyond reminders
Track outcomes and update dashboards	Sign off model updates before release
Raise alerts when accuracy or fairness drifts	Run compliance reviews and audits

Autonomy grows in phases

Phase 1: AI recommends, humans approve

Phase 2: Reminders automated

Phase 3: Expand after validation

Responsible AI Guardrails

Fairness

- Age and Location excluded from the model; used only for monitoring
- Monthly check of recall and flag rate across groups; pause if the gap is 10 points or more
- Test card type and employment as possible proxies

Explainability

- Every score shows its top reasons in plain language
- A score is a likelihood, not a verdict: it guides who to contact first
- All decisions logged for review

Compliance and privacy

- Aligned to standards such as ECOA, GDPR and local collection rules, with legal review
- Only needed data used; raw records and IDs kept out of GenAI tools
- No limit cuts or rate rises from the score alone

Human oversight

- Humans approve sensitive actions and handle disputes
- Named owner in Collections for results and exceptions
- Customers can dispute an outcome and get a human review

Expected Business Impact

Business KPIs (quantitative)

- 30+ day delinquency rate: goal of 10% below the control group
- Cure and repayment rate on contacted accounts
- Cost per account collected and agent hours per case
- Payment-plan uptake and completion rate
- Model quality: AUC, recall and fairness gap between groups

Customer outcomes (qualitative)

- Earlier, supportive contact before payments are missed
- Clear options and plain-language reasons
- Fair treatment across customer groups
- Fewer unnecessary calls and less friction
- Scalable across the full portfolio

Targets, not forecasts: to be measured in an 8-week pilot with a control group. The current model is not yet validated (AUC about 0.5), so no savings figure is claimed.

Roadmap and Next Steps

1

Fix and validate

Before launch

- Confirm delinquency and Missed_Payments definitions with Geldium
- Fix payment-history conflicts (68 inconsistent records)
- Validate the model; AI recommends, humans approve

2

Pilot

8 weeks

- Customers with utilization above 50%
- Automated reminders plus a payment-plan offer
- Random control group and weekly fairness check

3

Scale

If validated

- Extend to more segments; automate low-risk actions
- Retrain regularly; audit fairness monthly
- Collections and Risk review results before each expansion